Tutorial Problem Set IV

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During this tutorial, we first had a short mini-quiz on some mathematical errors from Quiz 1. Thereafter, we solved problems related to limits.

1 Errors from Quiz 1

All these questions are based on errors that you, as a group, made on Quiz 1.

Problem 1. If x and y are real numbers such that $x^2 \geq y^2$, then what can we deduce?

(A) $x \geq y$ or $x \leq -y$

(B) $x \geq y$

(C) $|x| \geq |y|$

(D) $x \geq \pm y$

Problem 2. Which of the following is true for all real numbers x and y?

(A) $\sqrt{x^2 + y^2} = x + y$

(B) $\sin^{-1}(x) = \frac{1}{\sin(x)}$

(C) $x^2 y^3 = xy^5$

(D) $-(x-y)^2 = -x^2 + 2xy - y^2$

Problem 3. Which of the following statements are true?

(A) If $f : \mathbb{R} \to \mathbb{R}$ is a function, which has an inverse g on a restricted domain, then the natural domain of g is equal to the range of f.

(B) The sum of two odd functions is an odd function.

(C) If $f(x)$ is a polynomial of odd degree, then f is an odd function.

(D) Every one-to-one function $f : \mathbb{R} \to \mathbb{R}$ is either strictly increasing or strictly decreasing.

2 Basic problems

Problem 4. Compute the following limits, if they exist.

(a) $\lim_{x \to 3} \frac{9-x^2}{x^3-27}$

(b) $\lim_{x \to 0} \sin\left(\frac{1}{x}\right)$

(c) $\lim_{x \to 0} \frac{\sin(x)}{x^2}$

(d) $\lim_{x \to 0} \frac{(1+x)^{\frac{n}{2}} - (1-x)^{\frac{n}{2}}}{x}$

(e) $\lim_{x \to \frac{\pi}{4}} \frac{\sin(x) - \cos(x)}{x - \frac{\pi}{4}}$

(f) $\lim_{x \to 0} x \cdot \sin\left(\frac{1}{x}\right) = 0$

Problem 5. Determine all real numbers a and b such that

(a) $\lim_{x \to 2} \frac{x^2+ax+b}{x-2} = 5$

(b) $\lim_{x \to 0} \frac{x+a}{\sin(bx)} = 3$

3 Challenging problems

Problem 6. Consider the recursively defined sequence given by

$$x_1 = 1 \quad \text{and} \quad x_{n+1} = \sqrt{2+x_n} \quad \text{for } n \geq 1.$$

(a) In part (b) we will show that $\lim_{n \to \infty} x_n$ exists. Under this assumption, compute the limit.

(b) Prove that the limit in (a) exists. For this, you may use the following theorem without proof.

Theorem 1 (Monotone convergence)**.** *Let (a_n) be a sequence and suppose that there exists $M > 0$ such that for all $n \geq 1$:*

(i) $a_n \leq a_{n+1}$,

(ii) $a_n \leq M$.

Then, $\lim_{n \to \infty} a_n$ *exists.*